

Deriving the Relaxation Time Formula

Bridging Physics and Econometrics

This derivation elegantly connects continuous-time physical concepts (decay processes) with the discrete-time series models (econometrics) used in actual data analysis.

Step 1: System Recovery in Continuous Time (Physics Perspective)

Assume a system deviates from its equilibrium due to an external shock. Without new shocks, it naturally recovers. In classic models (like Newton's law of cooling or the Ornstein-Uhlenbeck process in finance), **the speed of recovery is proportional to the current deviation.**

Expressed as a differential equation, the rate of change of the deviation $y(t)$ is:

$$\frac{dy(t)}{dt} = -ky(t)$$

Here, $k > 0$ is the **decay rate**. The negative sign ensures the deviation shrinks toward 0.

Solving this first-order ODE gives the state at any time t :

$$y(t) = y_0 e^{-kt}$$

(Where y_0 is the initial deviation at the moment of the shock. This shows the shock decays exponentially.)

Step 2: The Rigorous Definition of "Relaxation Time"

In physics and engineering, we don't ask "when will the system reach exactly 0?" (as exponential functions only infinitely approach 0). Instead, we ask:

"How long does it take for the system to decay to $1/e$ (approx. 36.8%) of its initial shock scale?" This specific duration is defined as the **Relaxation Time (τ)**.

By definition, when $t = \tau$, $y(\tau) = y_0 e^{-1}$. Substituting this into our solution:

$$y_0 e^{-k\tau} = y_0 e^{-1}$$

Canceling y_0 and comparing the exponents:

$$k\tau = 1 \implies \tau = \frac{1}{k}$$

Conclusion: Relaxation time is the reciprocal of the decay rate k .

Step 3: Mapping Continuous to Discrete Time Series

In reality (e.g., with CPI or consumption data), we cannot observe data continuously. We sample at fixed intervals, let's call it Δt (e.g., monthly).

Looking at two points separated by Δt , based on the Step 1 solution:

$$y(t + \Delta t) = y(t)e^{-k\Delta t}$$

When we introduce the continuous random shocks of the real world (the noise term $\sigma\eta_t$), this becomes the standard **discrete AR(1) model**:

$$y_{t+1} = Cy_t + \sigma\eta_t$$

Comparing these two equations reveals the core relationship: the regression coefficient C is the specific manifestation of the continuous decay equation over interval Δt :

$$C = e^{-k\Delta t}$$

Step 4: Deriving the Final Formula

We now have the two necessary pieces:

1. Relaxation Time: $\tau = \frac{1}{k}$
2. Discrete Model Coefficient: $C = e^{-k\Delta t}$

We need to convert C and Δt (which you can calculate from data) into τ .

First, take the natural logarithm ($\ln$) of both sides of the coefficient equation:

$$\ln(C) = -k\Delta t$$

Next, solve for the unknown decay rate k :

$$k = -\frac{\ln(C)}{\Delta t}$$

Finally, substitute k into the definition of τ :

$$\tau = \frac{1}{-\frac{\ln(C)}{\Delta t}}$$

Summary & Application Notes

- The regression coefficient C acts as the system's "**retention rate**." * Because $0 < C < 1$, its natural logarithm $\ln(C)$ will always be a negative number. The negative sign in the formula cancels this out, ensuring τ is positive.
- **Practical shortcut:** If your data is monthly ($\Delta t = 1$), the formula simplifies nicely to:

$$\tau = -\frac{1}{\ln(C)}$$